

Metadefinition 1 (Formula). If a, b are variables, then $a \in b$ is a formula. If p is a formula, then $\neg p$ is a formula. If p, q are formulas, then $p \text{ AND } q$ is a formula. If x is a variable, and p is a formula, then $\exists x(p)$ is a formula.

Metadefinition 2. If $p(x)$ is a formula, and y is a variable, then $p(y)$ is $p(x)$ with each occurrence of x replaced with y .

Notation 1 (Disjunction).

$$p \text{ OR } q \equiv \neg(\neg p \text{ AND } \neg q)$$

Notation 2 (Implication).

$$p \Rightarrow q \equiv \neg(p \text{ AND } \neg q)$$

Notation 3 (Equivalence).

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \text{ AND } (q \Rightarrow p)$$

Notation 4 (Exclusive Disjunction).

$$p \text{ XOR } q \equiv \neg(p \Leftrightarrow q)$$

Notation 5 (Negated Existential Quantification).

$$\nexists \equiv \neg \exists$$

Notation 6 (Bounded Existential Quantification).

$$\exists x \in A : (p) \equiv \exists x : (x \in A \text{ AND } p)$$

Notation 7 (Universal Quantification).

$$\forall x : (p) \equiv \nexists x : (\neg p)$$

Notation 8 (Negated Universal Quantification).

$$\nforall \equiv \neg \forall$$

Notation 9 (Bounded Universal Quantification).

$$\forall x \in A : (p) \equiv \forall x : (x \in A \Rightarrow p)$$

Notation 10 (Equality).

$$X = Y \equiv \forall a : (a \in X \Leftrightarrow a \in Y) \text{ AND } \forall B : (X \in B \Leftrightarrow Y \in B)$$

Notation 11 (Negated Equality).

$$a \neq b \equiv \neg(a = b)$$

Notation 12 (Unique Quantification).

$$\exists!x : (p(x)) \equiv \exists x : (p(x) \text{ AND } \forall y : (p(y) \Rightarrow y = x))$$

Axiom 1 (Empty Set).

$$\exists n, \nexists x : (x \in n)$$

Definition 1 (Empty Set).

$$n = \emptyset \Leftrightarrow \nexists x : (x \in n)$$

Axiom 2 (Extensionality).

$$\forall A, \forall B : (\forall x : (x \in A \Leftrightarrow x \in B) \Rightarrow A = B)$$

Axiom 3 (Pairing).

$$\forall a, \forall b, \exists C, \forall c : (c \in C \Leftrightarrow (c = a \text{ OR } c = b))$$

Definition 2 (Couple).

$$C = \{a, b\} \Leftrightarrow \forall c : (c \in C \Leftrightarrow (c = a \text{ OR } c = b))$$

Notation 13 (Singleton).

$$\{a\} \equiv \{a, a\}$$

Axiom 4 (Union).

$$\forall B, \exists U, \forall c : (c \in U \Leftrightarrow \exists P \in B : (c \in P))$$

Definition 3 (Union).

$$U = \bigcup B \Leftrightarrow \forall c : (c \in U \Leftrightarrow \exists P \in B : (c \in P))$$

Definition 4 (Binary Union).

$$A \cup B = \bigcup \{A, B\}$$

Definition 5 (Intersection).

$$I = \bigcap B \Leftrightarrow B \neq \emptyset \text{ AND } \forall c : (c \in I \Leftrightarrow \forall P \in B : (c \in P))$$

Definition 6 (Binary Intersection).

$$A \cap B = \bigcap \{A, B\}$$

Axiom 5 (Power Set).

$$\forall X, \exists P, \forall S : (S \in P \Leftrightarrow \forall s \in S : (s \in X))$$

Definition 7 (Power Set).

$$P = \mathcal{P}(X) \Leftrightarrow \forall S : (S \in P \Leftrightarrow \forall s \in S : (s \in X))$$

Definition 8 (Subset).

$$S \subseteq X \Leftrightarrow S \in \mathcal{P}(X)$$

Definition 9 (Successor).

$$\text{successor}(x) = x \cup \{x\}$$

Definition 10 (Inductive Set).

$$\text{Inductive}(I) \Leftrightarrow \emptyset \in I \text{ AND } \forall i \in I : (\text{successor}(i) \in I)$$

Axiom 6 (Infinity).

$$\exists I : (\text{Inductive}(I))$$

Definition 11 (Pair).

$$(a, b) = \{\{a\}, \{a, b\}\}$$

Axiom 7 (Foundation).

$$\forall X : (X \neq \emptyset \Rightarrow \exists x \in X : (x \cap X = \emptyset))$$

Definition 12 (Set of Free Variables).

$$\begin{aligned}\text{free}(a \in b) &= \{a, b\} \\ \text{free}(\neg p) &= \text{free}(p) \\ \text{free}(p \text{ AND } q) &= \text{free}(p) \cup \text{free}(q) \\ \text{free}(\exists x : (p)) &= \text{free}(p) \setminus \{x\}\end{aligned}$$

Axiom 8 (Replacement). Let $B \notin \text{free}(p)$.

$$\forall A : (\forall x \in A, \exists! y : (p) \Rightarrow \exists B, \forall y : (y \in B \Leftrightarrow \exists x \in A : (p)))$$

Definition 13 (Set Builder). Let $X \notin \text{free}(p)$.

$$X = \{a \in A \mid p\} \Leftrightarrow \forall x : (x \in X \Leftrightarrow (x \in A \text{ AND } p))$$

Definition 14 (Cartesian Product).

$$A \times B = \{(a, b) \mid a \in A \text{ AND } b \in B\}$$

Definition 15 (Set of Functions).

$$Y^X = \{f \subseteq X \times Y \mid \forall x \in X, \exists! y \in Y : ((x, y) \in f)\}$$

Definition 16 (Function).

$$\begin{aligned}f : X \rightarrow Y &\Leftrightarrow f \in Y^X \\ f(x) = y &\Leftrightarrow (x, y) \in f\end{aligned}$$

Definition 17 (Indexed Product).

$$\prod B = \left\{ c : B \rightarrow \bigcup B \mid \forall P \in B : (c(P) \in P) \right\}$$

Axiom 9 (Choice).

$$\forall O : \left(\forall o \in O : (o \neq \emptyset) \Rightarrow \prod O \neq \emptyset \right)$$

Definition 18 (Exclusion).

$$A \setminus B = \{x \in A \mid x \notin B\}$$

Definition 19 (Set of Natural Numbers).

$$N = \mathbb{N} \Leftrightarrow \forall n : (n \in N \Leftrightarrow \forall I : (\text{Inductive}(I) \Rightarrow n \in I))$$

Definition 20 (Strict Order of Natural Numbers). Let $a, b \in \mathbb{N}$.

$$a < b \Leftrightarrow a \in b$$

Notation 14 (Non-Strict Order).

$$a \leq b \equiv a < b \text{ OR } a = b$$

Notation 15 (Numeral).

$$0 \equiv \emptyset$$

$$1 \equiv \{0\}$$

$$2 \equiv \{0, 1\}$$

$$3 \equiv \{0, 1, 2\}$$

$$4 \equiv \{0, 1, 2, 3\}$$

$$5 \equiv \{0, 1, 2, 3, 4\}$$

$$6 \equiv \{0, 1, 2, 3, 4, 5\}$$

$$7 \equiv \{0, 1, 2, 3, 4, 5, 6\}$$

$$8 \equiv \{0, 1, 2, 3, 4, 5, 6, 7\}$$

$$9 \equiv \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

Definition 21 (Tuple). Let $n \in \mathbb{N} \setminus \{0\}$.

$$(x_1, \dots, x_n) = \begin{cases} x_1, & \text{if } n = 1 \\ (x_1, x_2), & \text{if } n = 2 \\ ((x_1, \dots, x_{n-1}), x_n) & \text{otherwise} \end{cases}$$